

Qatar-1b

R-2630

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_140915 · created 2026-08-29T00:20:25+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456915.71423 +/- 0.00037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.018
Transit depth (Rp/Rs)^2	1.87 +/- 0.5 [%]
Orbital Inclination (inc)	83.81 +/- 1.34
Airmass coefficient 1 (a1)	0.919 +/- 0.01
Airmass coefficient 2 (a2)	0.066 +/- 0.033
Scatter in the residuals of the lightcurve fit is	1.13 %
Best Comparison Star	#2 - [486, 108]
Optimal Aperture	3.35
Optimal Annulus	7.35
Transit Duration (day)	0.0658 +/- 0.0083

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.137 ± 0.018 vs 0.1463 (deviation 0.52σ)
Duration fit vs published	0.0658 d vs 0.0692 d (deviation 0.41σ)
Mid-time O–C	-0.9 min
Depth SNR	7.6
Residual scatter	1.13 %

Light curve

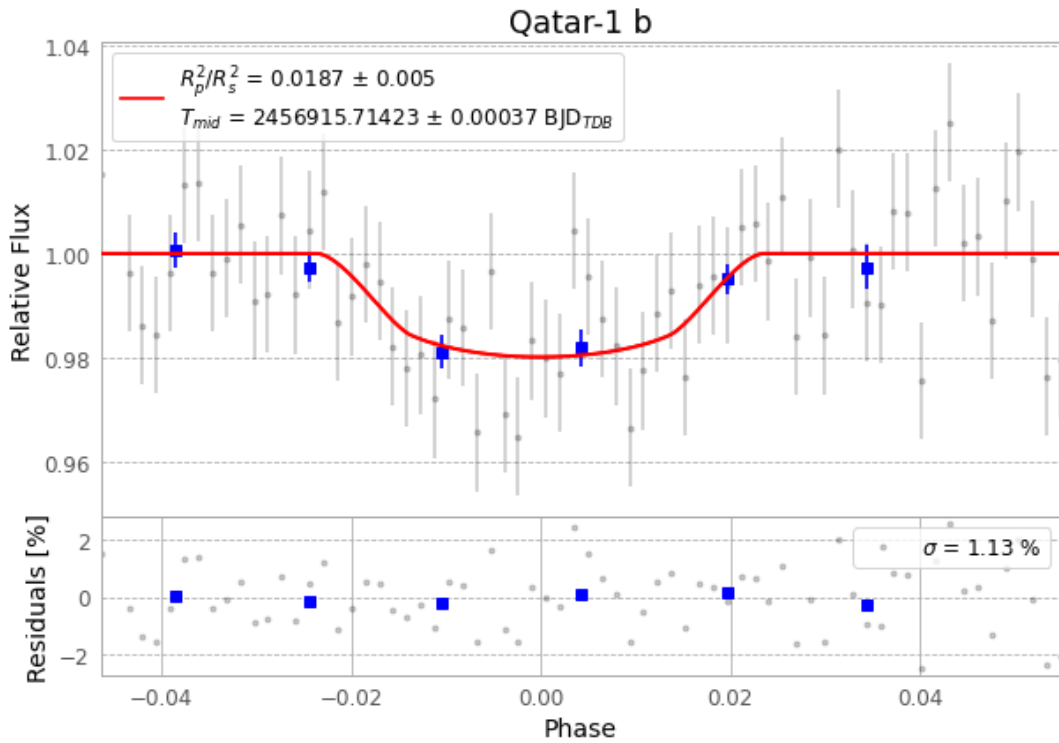


Figure 1 — FinalLightCurve_Qatar-1 b_2014-09-14.png (SHA-256 d60f6437a8e84b67...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [293, 264], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6dafa84ae17536ea...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), Qatar-1140915032412.FITS → Qatar-1140915065712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-140915.log (d34d372e9a2f...), FinalLightCurve_Qatar-1 b_2014-09-14.png (d60f6437a8e8...), FinalLightCurve_Qatar-1 b_2014-09-14.csv (256551a02178...)

Compute resources

Wall time 123.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 00:20Z.